

THE $K_3 \times E$ CY₃ PROGRAMME: UNIVERSAL κ_{BKM} AND THE ADVERSARIAL PENTAGON

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ABSTRACT. SCAFFOLD STUB. The CY₃ total space $K_3 \times E$ supports a chiral algebra $\mathcal{A}_{K_3 \times E}$ whose Borchers–Kac–Moody shadow satisfies $\kappa_{\text{BKM}}(\Phi_N) = c_N(\mathfrak{o})/2$ across the five Niemeier-lattice levels $N \in \{1, 2, 3, 4, 6\}$ (scope: five-level universal; $N \geq 7$ not covered here). The six candidate routes to the $K_3 \times E$ generator algebra (spacetime, worldsheet heterotic, F-theory, DT, BPS, Φ -functor) *do not* converge isomorphically (CY-C naive claim falsified; retraction B89): they form an adversarial *pentagon* of five intertwiners, with the R_2 source branch carrying the minimal rank. The pentagon stratifies by *generator rank* ρ^{R_i} , not by κ_{ch} -- the latter is a route-independent Hodge supertrace invariant equal to $\mathfrak{o}$ (retraction B90). The $N = 1$ coincidence $\kappa_{\text{BKM}} = \kappa_{\text{ch}} + \chi(\mathcal{O}_{\text{fiber}})$ fails for $N \geq 2$.

PRIMARY THEOREMS

- thm:borcherds-weight-kappa-BKM-universal ($\kappa_{\text{BKM}}(\Phi_N) = c_N(\mathfrak{o})/2$ universal).
- thm:pentagon-generator-rank-stratification (retraction of naive “six routes converge”; stratification by ρ^{R_i} , not κ_{ch}).
- $\kappa_{\text{ch}}(K_3 \times E) = \mathfrak{o}$ (Hodge supertrace, retraction B90).
- $N = 1$ coincidence failure for $N \geq 2$.

CHAPTER INPUTS (POINTERS)

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\input{../chapters/examples/k3e_cy3_programme.tex}
\input{../chapters/examples/k3e_bkm_chapter.tex}
\input{../chapters/examples/cy_c_six_routes_generator_level_platonic.tex}
\input{../chapters/examples/cy_c_pentagon_hypothesis_closures_platonic.tex}
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RETRACTIONS INSCRIBED

B89 (CY-C six-routes naive isomorphic convergence) and B90 (CY-C pentagon κ_{ch} stratification category error) retracted at primary source. See notes/rectification_map_beilinson_audit.md.